Project Brief

Dementia CARE (Community App for Resources)

Background

This project is inspired by the functionality and community-powered model of iNaturalist—a platform where users contribute, validate, and explore data collaboratively.

While iNaturalist focuses on mapping biodiversity, this project aims to apply a similar system to **map and share community-based resources for people living with dementia and their caregivers**, with potential to expand into broader accessibility and disability support.

Project Purpose

To design and develop a **user-friendly, community-driven mobile and/or web application** that allows users to:

- Discover dementia-friendly and accessibility-supportive resources in their area
- Contribute and validate information about services, spaces, and supports
- Build a shared, evolving map of inclusive community infrastructure

Problem Statement

Individuals living with dementia, and those who care for them, often face significant barriers in finding **reliable, up-to-date, and location-specific resources** such as:

- Safe public spaces
- Support services and programs
- Health and community services
- Dementia-friendly businesses

Currently, this information is **fragmented, difficult to navigate, or not centralized**, creating stress and limiting access to care and participation in community life.

Project Objectives

1. **Accessibility First**
   Design an intuitive interface tailored to:
   a. Caregivers
   b. Older adults
   c. People with cognitive impairments

2. **Community Contribution Model**
   Enable users to:
   a. Add new resources (e.g., locations, services)
   b. Tag and categorize entries
   c. Upload photos or notes
   d. Validate or review existing entries

3. **Geospatial Mapping**
   Provide a map-based interface where users can:
   a. Search nearby resources
   b. Filter by category (health, social, safety, etc.)
   c. View detailed resource profiles

4. **Trust & Verification**
   Incorporate a system for:
   a. Community validation (similar to iNaturalist confirmations)
   b. Moderation or trusted contributors

5. **Scalability**
   Design with future expansion in mind to include:
   a. Other disabilities (mobility, sensory, neurodiversity)
   b. Integration with existing service directories

Core Features (MVP – Minimum Viable Product)

- User account creation (simple, low-barrier)

- Map-based interface with geolocation
- Ability to:
    - Add a resource (name, type, description, location)
    - View and search resources
    - Filter by categories (e.g., dementia-friendly, accessible washrooms, respite services)
- Basic validation system (e.g., "verified by community")
- Simple rating or tagging system

Potential Advanced Features (Stretch Goals)

- AI-assisted tagging or categorization
- Accessibility scoring (user-generated)
- Saved routes or "safe paths"
- Offline functionality
- Integration with health or municipal datasets
- Multilingual support (important for inclusive access)

Design Principles

- **Simplicity over complexity**
- **High contrast, readable UI**
- **Minimal cognitive load**
- **Clear navigation and labeling**
- **Inclusive and culturally sensitive design**

Target Users

- Caregivers of people living with dementia
- Individuals living with early-stage dementia
- Families and support networks
- Service providers and community organizations

Expected Outcomes

- A working prototype or proof of concept
- UX/UI mockups or wireframes
- Defined technical architecture
- Recommendations for scaling and real-world deployment

Student Opportunity & Learning Goals

Students participating in this project will gain experience in:

- Human-centered and inclusive design
- Real-world problem solving in health and accessibility
- Geolocation and mapping technologies
- Collaborative product development
- Ethical and community-informed technology design

Collaboration & Support

You will be working with stakeholders who can provide:

- Context on dementia and accessibility needs
- User insights and testing feedback
- Strategic direction and real-world application context

App Development Roadmap

To support the practical development of the application within the context of a student-led capstone team, the development process should follow a modular path. Each stage will focus on achieving critical features and functionality that can be tested in real-world circumstances and environments.

This approach ensures that the project scope and deliverables at each milestone result in a usable, testable product without overcommitting the student team. The proposed modules support the initial framework and architectural creation, front-end development (including accessible UI/UX), and back-end coding and hosting implementation.

Depending on project team size modules can be allocated to multiple teams simultaneously or across multiple semester or course session.

Module 1: Technical Framework Specification and UI/UX Design

- Technical Architecture: Define the full technology stack and backend infrastructure.
- Requirement Analysis: Perform complete feature discovery and a formal Software Requirements Analysis (SRA).
- UI/UX Design: Develop fully scoped page views and interface elements.
- Accessibility: Implement advanced accessibility best practices and compliance standards.
- Front-End Development: Complete build-out of all user-facing features and interface components.

Final Deliverable: Proof of concept release and documentation package.

Module 2: Core Development

- Backend Development: Complete build-out of server-side logic and core functionality. Architecture & Hosting: Design of the server environment and selection of hosting infrastructure.
- Security & Privacy: Implementation of data protection protocols and privacy compliance.
- Scalable Storage: Setup of flexible data and file storage solutions.
- Integrations: Full integration of third-party products and GIS platform.

Final Deliverable: Functional and tested back end.

Module 3: Integration and Production Release

- System Integration: Full integration of the front-end with the back-end API and hosting infrastructure.
- Quality Assurance: Comprehensive software testing to ensure reliability and accessibility.
- Performance: Regular measurement of system speed, responsiveness, and load handling.
- Operations: Planning for ongoing maintenance, security updates, and long-term support.

Final Deliverable: Production release.

Key Question for the Team

How might we design a community-powered mapping tool that makes invisible support systems visible, accessible, and trustworthy for those navigating dementia and disability?